

Assignment 3_Claxton_funds of ML

2026-06-10

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.5.3
## Loading required package: ggplot2
## Warning: package 'ggplot2' was built under R version 4.5.3
## Loading required package: lattice
```

```
library(reshape)
```

```
## Warning: package 'reshape' was built under R version 4.5.3
```

```
library(e1071)
```

```
## Warning: package 'e1071' was built under R version 4.5.3
##
## Attaching package: 'e1071'
##
## The following object is masked from 'package:ggplot2':
##
##     element
```

```
library(gmodels)
```

```
## Warning: package 'gmodels' was built under R version 4.5.3
```

```
bank.df<-read.csv("C:\\Users\\User\\OneDrive\\Desktop\\Grad School_Business Analytics\\Fundamentals of ML\\data\\bank.csv")
#read CSV and save as dataframe

bank.df <- bank.df[, c("Online", "CreditCard", "Personal.Loan")]
#reduce variables to Online, CC, and Personal.Loan

bank.df$Online <- factor(bank.df $Online, levels = c(0,1))

bank.df$CreditCard <- factor(bank.df $CreditCard, levels = c(0,1))

bank.df$Personal.Loan <- factor(bank.df $Personal.Loan, levels = c(0,1))
#convert to factors

set.seed(123)
#set seed
```

```

# Stratified partition: 60% training

train_index <- createDataPartition(bank.df $Personal.Loan, p = 0.6, list = FALSE)

train <- bank.df [train_index, ]

valid <- bank.df [-train_index, ]

train$dummy <- 1

train_melt <- melt(train,

                    id.vars = c("CreditCard", "Online", "Personal.Loan"),

                    measure.vars = "dummy")
#create data partition, establish train and test
#use melt() to convert data into long form
pivot_A <- cast(train_melt,
                CreditCard + Personal.Loan ~ Online,
                fun.aggregate = sum,
                value = "value")
#Use case to create the pivot table.
#Aggregate and value are just counting rows
pivot_A

```

```

##   CreditCard Personal.Loan   0   1
## 1          0             0 791 1144
## 2          0             1  79  125
## 3          1             0 310  467
## 4          1             1  33   51

```

Part A: This is the pivot table with credit card and personal loan on the rows with online as a column variable Part B: Consider a customer who owns a credit card and uses online banking. Looking at the pivot table, the probability of this customer accepting a private loan (based on the training data) is $51/(467+51) = 0.098$.

```

# (Part C) Pivot tables for Loan as a function of Online

pivot_LoansByOnline <- table(train$Personal.Loan, train$Online)

pivot_LoansByOnline

##
##      0      1
## 0 1101 1611
## 1  112  176

```

```

# Pivot table for loan as a function of CC

pivot_LoansByCC <- table(train$Personal.Loan, train$CreditCard)

pivot_LoansByCC

```

```
##
##      0      1
##  0 1935  777
##  1  204   84
```

Part C: These two pivot tables depict (from top to bottom) Loan as a function of Online status and Loan as a function of Credit Card status.

Part D: $P(\text{CreditCard} = 1 \mid \text{Loan} = 1) = 0.29$ $P(\text{Online}=1 \mid \text{Loan}=1) = 0.61$ $P(\text{Loan} = 1) = 0.10$
 $P(\text{CreditCard}=1 \mid \text{Loan} = 0) = 0.29$ $P(\text{Online}=1 \mid \text{Loan} = 0) = .59$ $P(\text{Loan}=0) = .90$

Part E: Compute the Naive Bayes probability = Probability of accepting a loan when the user has a credit card and uses online banking -> $P(\text{Loan}=1 \mid \text{CreditCard}=1, \text{Online}=1) \rightarrow \text{Loan} = 1 / \text{Loan}=0 + \text{Loan} = 1$
 -> $.29 * .61 * .1 / ((.29 * .61 * .1) + (.29 * .59 * .90)) = 0.103$

Part G: What components do we need? We need all of the components of the table because we need the probabilities to compute Naive Bayes

```
nb_model <- naiveBayes(Personal.Loan ~ CreditCard + Online, data = train)

nb_model
```

```
##
## Naive Bayes Classifier for Discrete Predictors
##
## Call:
## naiveBayes.default(x = X, y = Y, laplace = laplace)
##
## A-priori probabilities:
## Y
##      0      1
## 0.904 0.096
##
## Conditional probabilities:
##      CreditCard
## Y      0      1
##  0 0.7134956 0.2865044
##  1 0.7083333 0.2916667
##
##      Online
## Y      0      1
##  0 0.4059735 0.5940265
##  1 0.3888889 0.6111111
```

```
#Conditional probabilities for Online | Personal.Loan and CC
# | Personal.Loan should match the answers from part D
```

```
new_customer <- data.frame(

  CreditCard = factor(1, levels = c(0,1)),

  Online = factor(1, levels = c(0,1)))

#create the new customer
```

```
pred_probs <- predict(nb_model, new_customer, type = "raw")
```

```
pred_probs
```

```
##           0           1  
## [1,] 0.8999139 0.1000861
```

```
#predict the class of the new customer  
#using the naiveBayes model
```

```
#This should match part E and NOT part B
```

This matches our calculations for part E because we manually calculated probabilities using Naive Bayes methodology. It does *NOT match exactly to part B because those probabilities were derived from empirical observations from the training data.